

Review of the Innate Immune Response in Acne vulgaris: Activation of Toll-Like Receptor 2 in Acne Triggers Inflammatory Cytokine Responses

Jenny Kim

Division of Dermatology, Department of Medicine, David Geffen School of Medicine at University of California Los Angeles, Los Angeles, Calif., USA

Key Words

Innate immunity · Toll-like receptors · Monocytes/macrophages · Acne, inflammatory cytokine responses

Abstract

Acne vulgaris is a common disorder that affects 40–50 million people in the USA alone. The pathogenesis of acne is multifactorial, including hormonal, microbiological and immunological mechanisms. One of the factors that contributes to the pathogenesis of acne is *Propionibacterium acnes*; yet, the molecular mechanism by which *P. acnes* induces inflammation is not known. Recent studies have demonstrated that microbial agents trigger cytokine responses via Toll-like receptors (TLRs). TLRs are pattern recognition receptors that recognize pathogen-associated molecular patterns conserved among microorganisms and elicit immune responses. We investigated whether TLR2 mediates *P. acnes*-induced cytokine production in acne. Using transfectant cells we found that TLR2 was sufficient for NF- κ B activation in response to *P. acnes*. In addition, peritoneal macrophages from wild-type, TLR6 knockout and TLR1 knockout mice, but not TLR2 knockout mice, produced IL-6 in response to *P. acnes*. *P. acnes* induced activation of IL-12 and IL-8 production by primary human monocytes, and this cytokine production was inhibited by anti-TLR2-blocking antibody. Finally, in acne lesions, TLR2 was expressed on the cell surface of macrophages sur-

rounding pilosebaceous follicles. These data suggest that *P. acnes* triggers inflammatory cytokine responses in acne by activation of TLR2. As such, TLR2 may provide a novel target for the treatment of this common skin disease.

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Introduction

Pathogenesis of Acne

Acne vulgaris is a common disorder that affects 40–50 million people in the USA alone. Although acne is rarely life threatening, it is a disease that can have a significant effect on the patients' physical and psychological well-being. The pathogenesis of acne is multifactorial, including hormonal, microbiological and immunological mechanisms.

Acne lesions first develop in the pilosebaceous unit as subclinical microcomedones that may evolve into closed or open comedones and inflammatory lesions such as papules, pustules, nodules and cysts. The formation of comedones is believed to involve hyperkeratinization of the follicle secondary to both hyperproliferation of keratinocytes lining the follicle as well as their reduced desquamation. The inflammatory cytokine IL-1 α has also been implicated in this process as IL-1 α has been shown to induce hyperkeratinization in vitro [1]. Microcomedones or comedones may then later develop into inflamma-

tory lesions as a result of CD4⁺ T-cell activation and migration [2], cytokine production by keratinocytes as well as macrophages and neutrophils recruited to the site, hormonal factors and enhanced sebum production.

One of the factors that contributes to the pathogenesis of acne is *Propionibacterium acnes*, part of the normal skin flora, but which can be significantly increased in pilosebaceous units of patients with acne [3]. This review will focus on how *P. acnes* elicits innate immune responses and plays a role in the pathogenesis of acne.

Propionibacterium acnes

P. acnes has long been implicated as an etiological factor of acne since it was demonstrated that its injection into the skin produces potent inflammatory responses [4]. This hypothesis was further supported by studies demonstrating failure of erythromycin therapy for acne in patients with erythromycin-resistant *P. acnes*, and studies of teenagers with acne demonstrating significantly higher numbers of *P. acnes* on their skin when compared to age-matched controls without acne [3, 5, 6].

Although *P. acnes* is a gram-positive bacterium, it is variably and weakly gram-positive. It is described as diphtheroid or coryneform because it is rod shaped and slightly curved. A number of unique features of the *P. acnes* cell wall and outer envelope further distinguish it from other gram-positive bacteria [7]. *P. acnes* synthesizes phosphatidyl inositol; this is unlike almost all other bacteria, but is made by virtually all eukaryotes. The peptidoglycan of *P. acnes* is distinct from most gram-positive bacteria, containing a cross-linkage region of peptide chains with *L,L*-diaminopimelic acid and *D*-alanine in which two glycine residues combine with amino and carboxyl groups. These structures may allow for the recognition of *P. acnes* by innate cells via pattern recognition receptors.

The role of *P. acnes* as an etiological factor in acne has been well established; however, the precise role that this bacterium plays in the pathogenesis of acne remains unclear. *P. acnes* contributes to the inflammatory nature of acne by inducing monocytes to secrete proinflammatory cytokines including TNF- α , IL-1 β and IL-8 [8]. In particular, IL-8 along with other *P. acnes*-induced chemotactic factors may play an important role in attracting neutrophils to the pilosebaceous unit. In addition, *P. acnes* releases lipases, proteases and hyaluronidases which contribute to tissue injury [9–12].

The mechanism by which *P. acnes* activates monocyte cytokine release was unknown but thought to involve pattern recognition receptors of the innate immune system

[8]. Recently identified Toll-like receptors (TLRs) are one example of pattern recognition receptor.

Toll-Like Receptors

Toll receptors were first identified in *Drosophila*, and mammalian homologs were found to mediate the immune response to microbial ligands [13, 14]. TLRs are transmembrane proteins capable of mediating responses to pathogen-associated molecular patterns conserved among microorganisms. The extracellular portion of TLRs is composed of leucine-rich repeats while the intracellular portion shares homology with the cytoplasmic domain of the IL-1 receptor. When TLRs are activated by exposure to microbial ligands, the intracellular domain of the TLR may trigger a MyD88-dependent pathway involving factors such as IL-1-receptor-associated kinase and tumor-necrosis-factor-receptor-activated factor 6 that ultimately lead to the nuclear translocation of the transcription factor NF- κ B. NF- κ B then acts to modulate expression of many immune response genes [15, 16]. However, ligand exposure may also trigger MyD88-independent pathways in some TLRs. There is evidence that TLR3 and TLR4 also trigger a MyD88-independent pathway that involves IRF-3 to ultimately activate NF- κ B [17].

Currently, 10 TLRs have been identified in humans (fig. 1). The microbial ligands for many of these receptors have been demonstrated and include bacterial cell wall components and genetic material. More specifically, TLR4 mediates host responses to bacterial lipopolysaccharide from gram-negative bacteria, TLR2 mediates responses to peptidoglycan from gram-positive bacteria, and TLR5 mediates the host response to bacterial flagellin [18–20]. In addition, TLR9 mediates the response to the unmethylated CpG DNA comprising bacterial genomes, and TLR3 mediates responses to viral double-stranded RNA [21, 22]. Furthermore, TLR heterodimers have been shown to mediate responses to microbial lipoproteins, with TLR2/1 heterodimers mediating responses to triacylated lipoproteins and TLR2/6 heterodimers mediating responses to diacylated lipoproteins [23, 24].

We devised a study to elucidate the mechanism by which *P. acnes* induces inflammatory cytokines in monocytes. The following section provides a summary of our previously published data [25] that *P. acnes* induces monocyte cytokine production through a TLR2-dependent pathway and that the expression of TLR2 in acne lesions indicates that activation of TLR2 can contribute to inflammation at the site of disease activity.

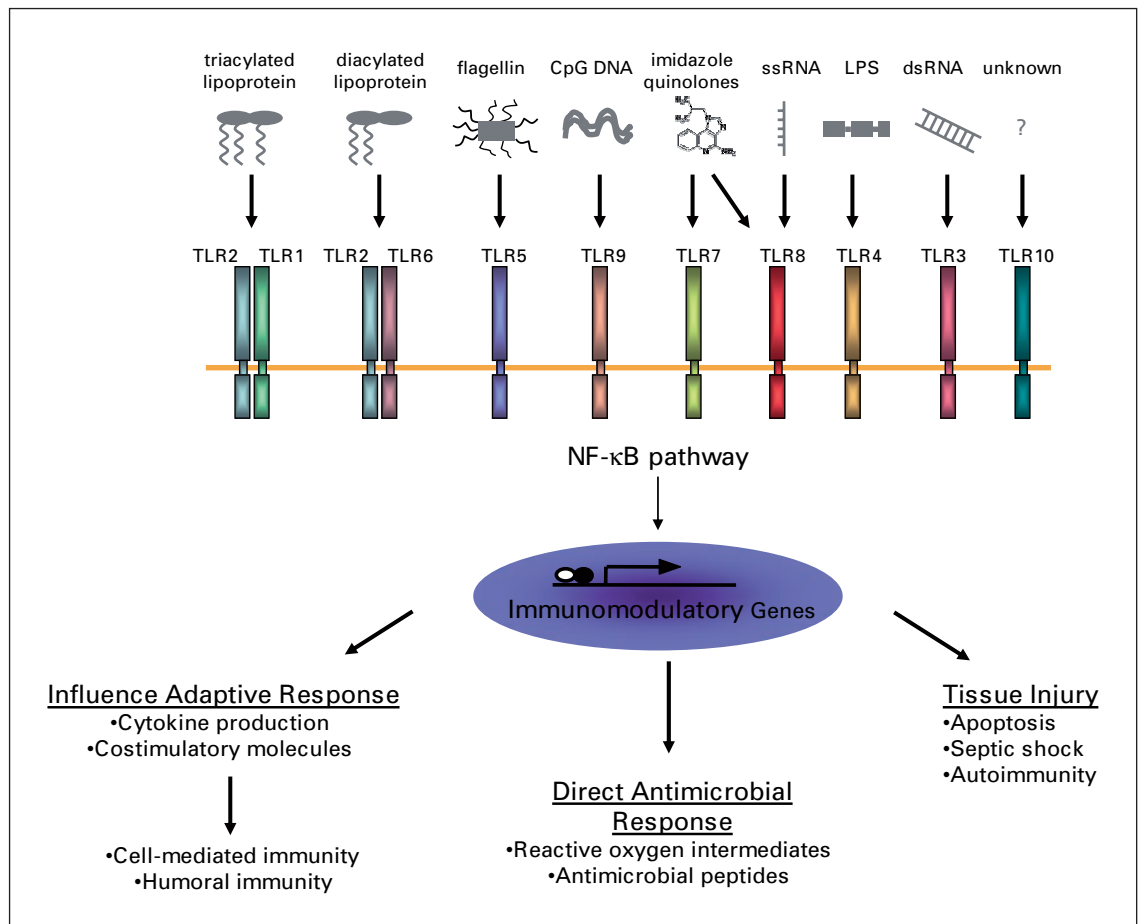


Fig. 1. Human TLRs and their ligands. CpG = Cytosine guanine nucleotides; LPS = lipopolysaccharide.

Results

TLR2 Is Sufficient for P. acnes Activation of Monocytes

Previous studies have demonstrated that TLR2 mediates the response of several ligands from gram-positive organisms. We therefore sought to determine whether TLR2 is sufficient for *P. acnes*-induced gene activation. We used TLR2 and TLR4 transfectant cells to measure the NF- κ B activation with *P. acnes*. We found that *P. acnes* induced NF- κ B in cells expressing TLR2 but not TLR4.

Since TLR2 cooperates with TLR6 and TLR1 in the recognition of microbial ligands [24, 26, 27], we examined the role of TLR6 and TLR1 in mediating activation by *P. acnes* using TLR knockout mice. Peritoneal macrophages were obtained from wild-type, TLR2 $^{-/-}$, TLR6 $^{-/-}$ and TLR1 $^{-/-}$ mice, activated with *P. acnes*,

and supernatants assayed for the presence of the inflammatory cytokine IL-6. *P. acnes* induced IL-6 release from wild-type mice, TLR1 $^{-/-}$ mice and TLR6 $^{-/-}$ mice but not TLR2 $^{-/-}$ mice (data not shown). These results indicate the specificity of the response that TLR2 but not TLR4, TLR6 or TLR1 mediates *P. acnes*-induced cell activation.

P. acnes Induces IL-12 and IL-8 Production by Human Adherent Monocytes

IL-12 is a pivotal cytokine in activating Th1 T-cell responses and is one of the major proinflammatory cytokines produced by monocytes in response to gram-positive organisms. To determine whether *P. acnes* induced IL-12 production via a TLR-dependent pathway, primary human monocytes from normal donors were stimulated with various dilutions of *P. acnes* sonicate, and cytokine production was measured. We found that *P. acnes*

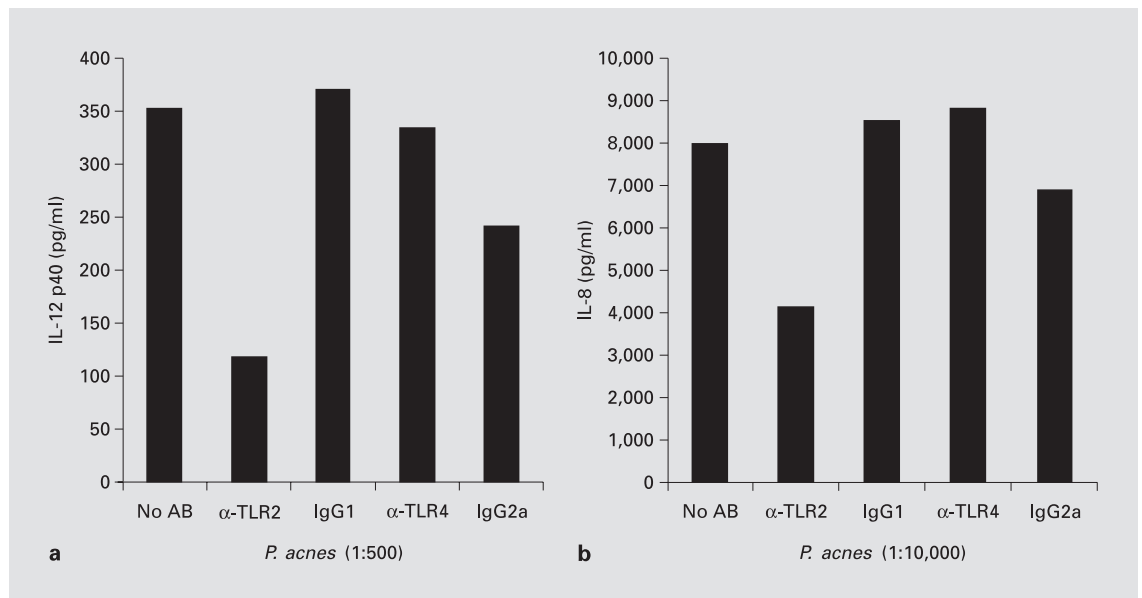


Fig. 2. *P. acnes* induces IL-12 and IL-8 production in human monocytes via TLR2. Primary human monocytes were incubated with anti-TLR2, anti-TLR4 or isotype control monoclonal antibodies (AB), then stimulated with *P. acnes*. The production of IL-12 p40 (**a**) and IL-8 (**b**) was determined by ELISA. Copyright 2002, American Association of Immunologists Inc.

induced IL-12 production by monocytes in a dose-dependent manner. *P. acnes* also induced the release of IL-8, a cytokine involved in neutrophil chemotaxis.

To determine whether the production of proinflammatory cytokines could be mediated through TLRs, monocytes from normal donors were cultured with anti-TLR2 and anti-TLR4 antibodies prior to stimulation with *P. acnes* sonicate. *P. acnes* induction of IL-12 production in monocytes was blocked by approximately 65% with the addition of anti-TLR2 antibody to the culture (fig. 2a). In contrast, the addition of anti-TLR4 and isotype control antibodies had no significant effect on IL-12 production by human monocytes, suggesting that the blocking effect was specific to TLR2. Similarly, the production of IL-8 was also blocked by approximately 50% by the addition of anti-TLR2 antibody (fig. 2b). Anti-TLR4 and isotype control antibodies had no significant effect on IL-8 production. These results suggest that one mechanism by which *P. acnes* induces IL-12 and IL-8 production in human monocytes is by the activation of TLR2.

TLR2 Is Expressed on Macrophages in Acne Lesions

We next wanted to determine whether there was any evidence for a TLR-dependent mechanism occurring at the site of the disease activity. We obtained acne biopsies

from patients and analyzed the lesions for TLR expression. Immunohistochemistry labeling using a monoclonal antibody specific to TLR2 revealed TLR2 expression on large ovoid cells within acne lesions. TLR2⁺ cells were detected primarily in the inflammatory infiltrate around the perifollicular/peribulbar region. Interestingly we found that the frequency of TLR2-expressing cells was determined according to the duration of the acne lesion. In early acne lesions, up to 6 h, few TLR2⁺ cells were detected (fig. 3). In lesions obtained between 12 and 24 h, TLR2⁺ cells were found to be more numerous around the pilosebaceous follicles. Finally, in older lesions obtained between 48 and 72 h, even greater numbers of TLR2⁺ cells were detected. From these experiments, we concluded that the infiltration of TLR2⁺ cells was an early event in the evolution of acne lesions. Furthermore, the frequency of cells expressing TLR2 proteins is upregulated during the inflammatory process up to 48–72 h after the onset of acne lesions.

To identify the lineage of cells expressing TLR2 in acne lesions, we performed a double immunofluorescence labeling and utilized confocal laser microscopy. The colocalization of TLR2 with CD14 suggests that TLR2 is expressed on cells of the monocyte/macrophage lineage in acne lesions (data not shown).

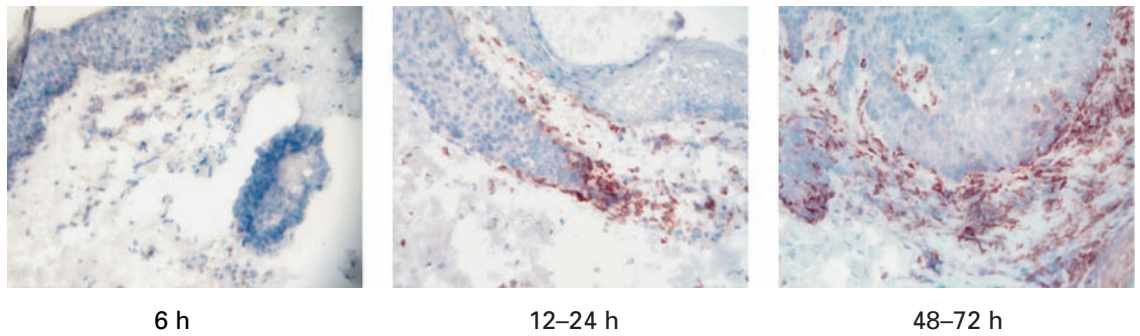


Fig. 3. TLR2 expression in lesions during various stages of the acne. Skin biopsy specimens of acne lesions were stained by the immunoperoxidase method with monoclonal antibodies specific for TLR2. In early acne lesions, up to 6 h, few TLR2+ cells were detected. In lesions obtained between 12 and 24 h, TLR2+ cells were much more numerous around the pilosebaceous follicles. In later older lesions obtained between 48 and 72 h, greater numbers of TLR2+ cells were detected. Copyright 2002, American Association of Immunologists Inc.

Discussion

Recognition of microbial pathogens by the cells of the immune system triggers host defense mechanisms to combat infection and prevent disease. However, activation of these same pathways can also result in inflammation at the site of disease and subsequent tissue injury. In acne, the host response to *P. acnes* can result in the production of proinflammatory cytokines and contribute to the clinical manifestations of disease. In our study, we investigated the molecular mechanism by which *P. acnes* induces proinflammatory cytokines in monocytes and provide evidence that the *P. acnes*-induced release of cytokines is dependent on TLR2. Furthermore, TLR2+ macrophages were present in acne lesions, infiltrating around pilosebaceous follicles and increased during the evolution of the disease. Our data suggest a novel mechanism by which *P. acnes* induces inflammation in acne by the activation of TLR2 and the subsequent release of cytokines which regulate the local immune response.

Currently the exact *P. acnes* TLR2 ligand has not been identified. However, our data [25] as well as findings from previous studies [8] point to *P. acnes* peptidoglycan on the cell wall as the TLR2 ligand. Further studies will be required to identify the TLR2 ligand present in *P. acnes* and to define its role in the pathogenesis of acne.

The primary event in inflammatory acne involves the disruption of the follicular epithelium and colonization of the follicles with *P. acnes* with subsequent inflammatory reactions in the surrounding dermis. The detection of TLR2+ cells in the perifollicular region provides indirect evidence that TLR2 activation contributes to the patho-

genesis of acne, suggesting that these cells promote inflammatory responses at the site of the disease activity. Furthermore, activation of TLR2 on monocytes releases proinflammatory cytokines, IL-12 and IL-8. IL-8 attracts neutrophils to the site of active lesion, and release of lysosomal enzymes by neutrophils leads to rupture of follicular epithelium and further inflammation [28]. On the other hand, IL-12 promotes development of Th1-mediated immune responses. Overproduction of Th1 cytokines such as IL-12 has been implicated in the development of tissue injury in certain autoimmune and inflammatory diseases [29–35]. In this manner, the activation of TLR2 on monocytes and other TLRs as well as other inflammatory cells is likely involved in the pathogenesis of acne. Given these data, TLR2 appears to be a logical target for therapeutic intervention in order to block inflammatory cytokine responses in acne and other inflammatory conditions in which tissue injury is detrimental to the host.

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